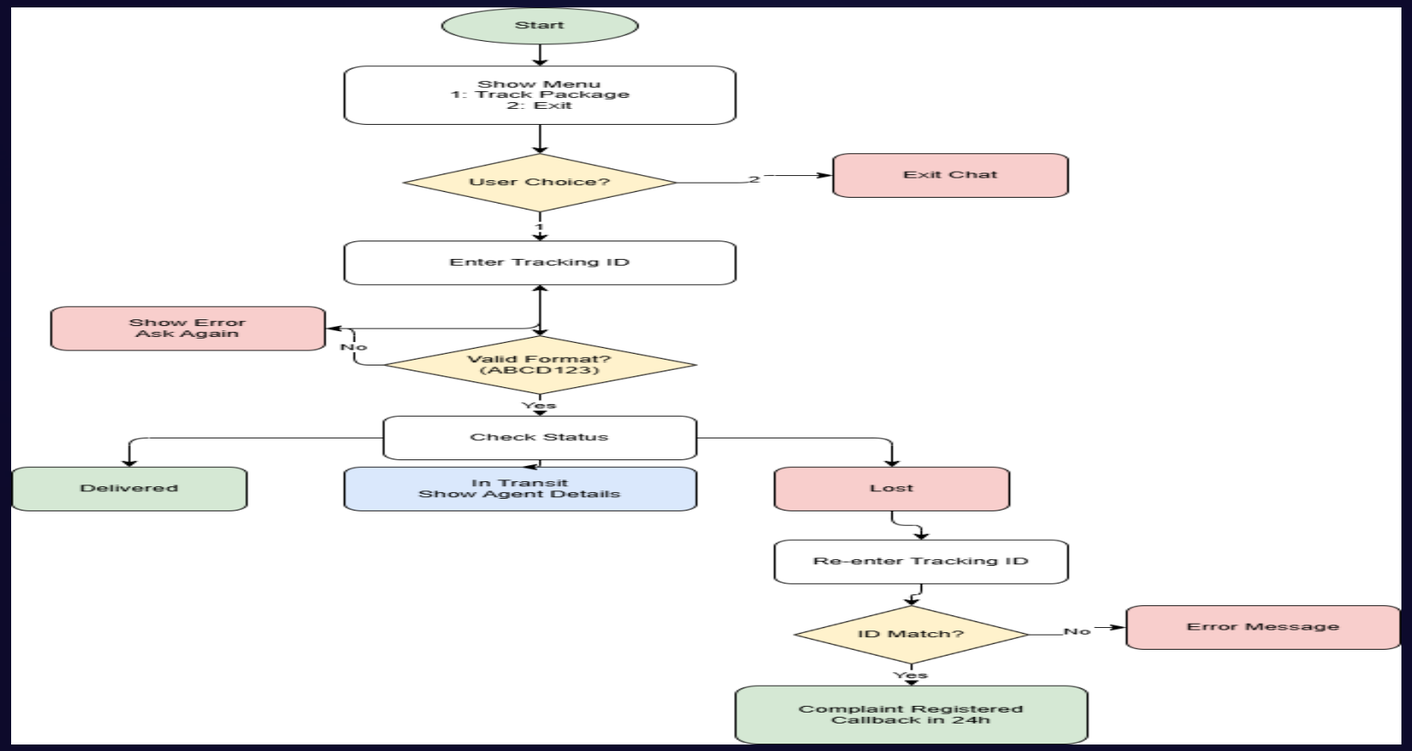
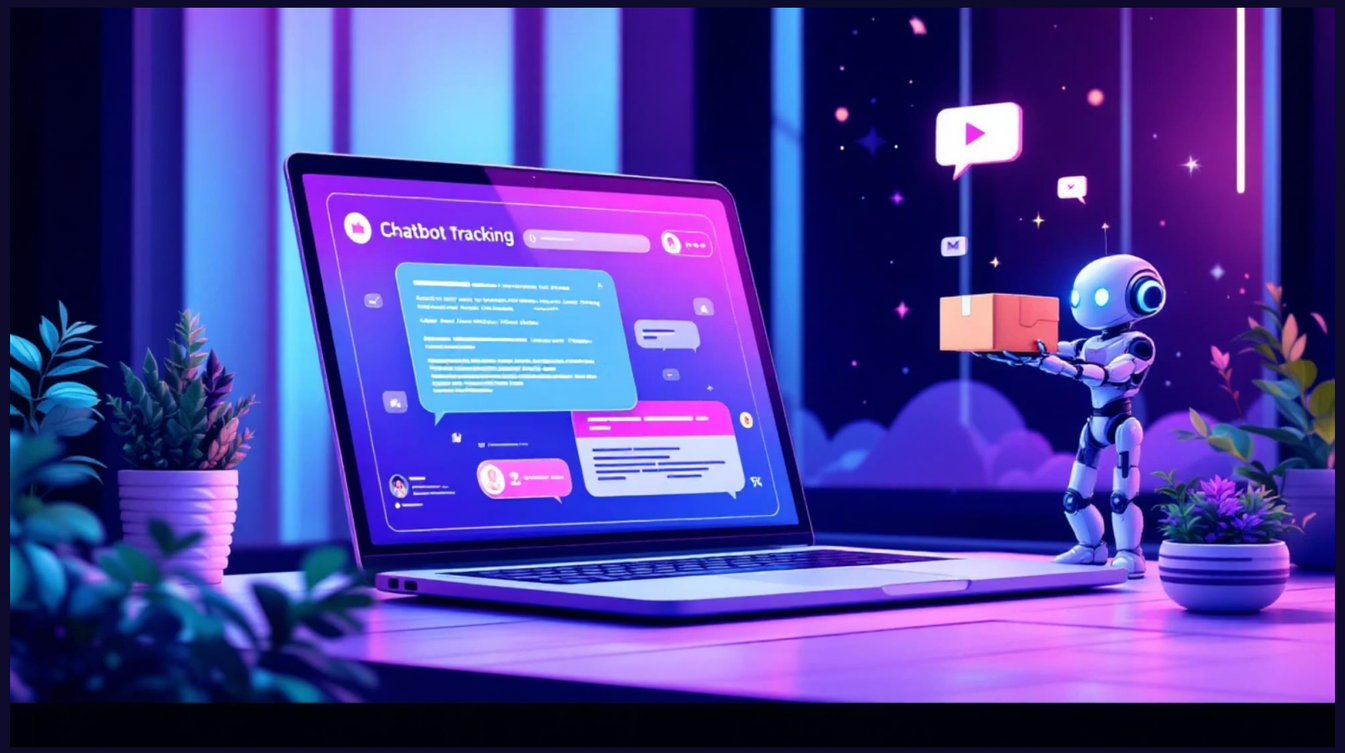
Conversation Design

FLOWCHART INCLUDED

A decision-tree guides users from greeting to resolution with guardrails for accuracy and clarity.



Logic ensures: option selection → input → validation → status handling → complaint registration with 24h callback assurance.



Prototype Implementation

1

Tech Stack

- Python CLI chatbot
- HTML, CSS, JavaScript web UI

2

Core Features

- Menu-driven navigation
- Regex/conditional validation
- Status simulation: Delivered/Transit/Lost
- Random agent name + contact
- Exit option anytime

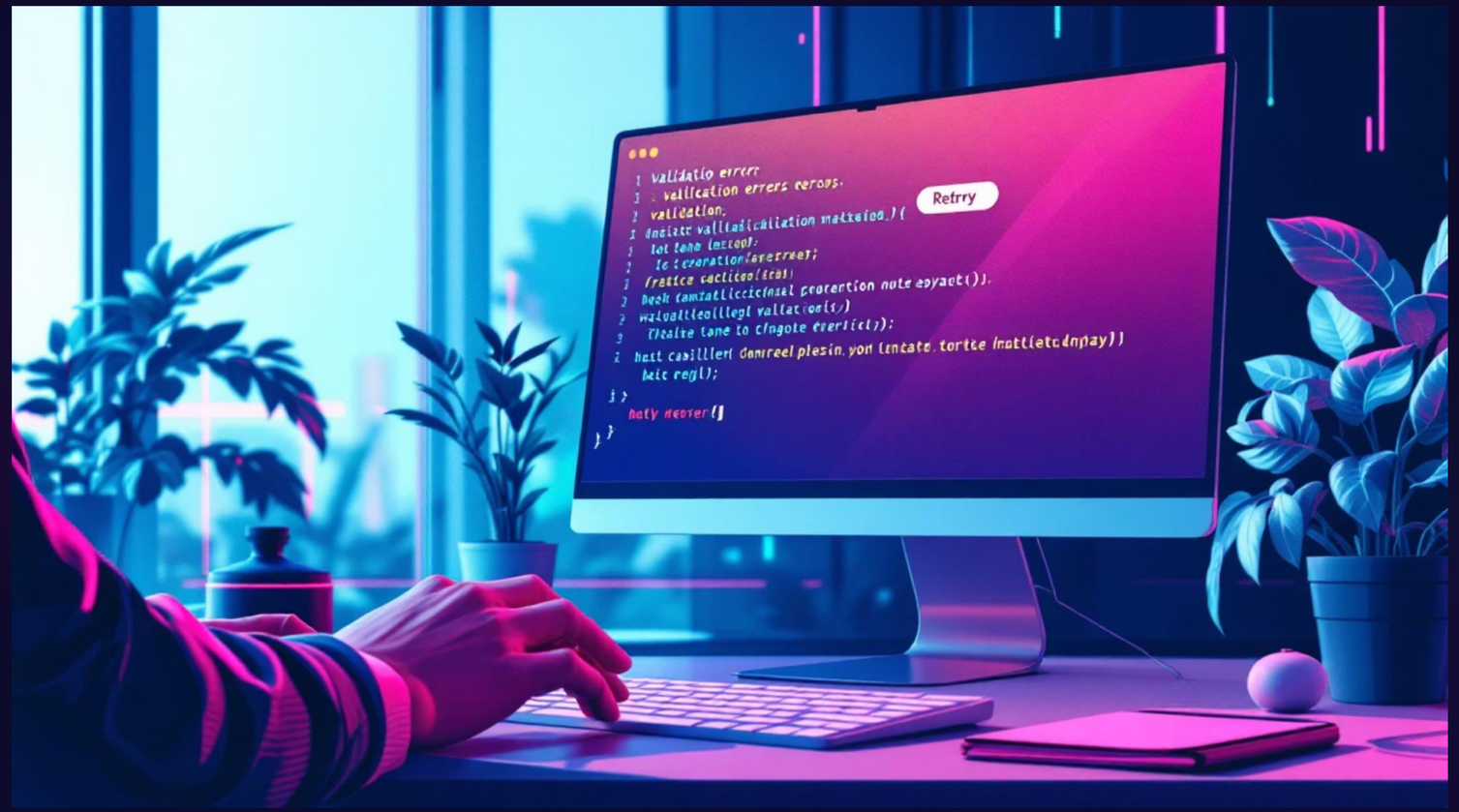
Error Handling

Example 1: Invalid Tracking ID

When format ≠ 4 letters + 3 digits (e.g., ABCD123), system displays an error and reprompts.

Example 2: Complaint Mismatch

If Lost flow ID re-entry doesn't match, system shows: "Tracking ID does not match".



Robust validation + clear guidance reduce frustration and prevent dead-ends.

Challenges Faced

Input Validation Complexity

- Ensuring tracking ID follows correct format (4 letters + 3 digits)
- Rejecting invalid inputs

Handling Unexpected User Inputs

- Managing incorrect entries and empty inputs
- Handling mismatched tracking IDs during complaint registration

Designing Conversation Flow

- Structuring logical and user-friendly chatbot flow
- Creating clear decision paths

Simulating Realistic Data

- Generating believable package statuses (Delivered, In Transit, Lost)
- Creating realistic delivery agent details

Frontend-Backend Integration

- Connecting HTML/CSS/JavaScript interface with Python backend
- Handling communication issues

Error Handling Implementation

- Providing meaningful error messages
- Maintaining smooth user experience

Design Choices, Challenges & Improvements

Design Choices

- Simple menu UI
- Stepwise decision tree
- Strict input validation

Challenges

- Managing bad inputs
- Realistic conversation paths
- Frontend–backend syncing

Improvements

- Database-backed tracking
- Live shipment APIs
- AI/NLP for natural chat
- Refined UI/UX

Next step: connect to carrier APIs and log interactions for analytics-driven optimization.